Predicting Ridership at the Chicago Transit Authority's 'L' Stations

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Introduction

The Chicago Transit Authority (CTA) serves the third most populous and second densest city in the United States by moving hundreds of thousands of travelers every day. According to the U.S. Census Bureau's 2024 American Community Survey, 20.5% of Chicago residents rely on public transportation to get to work. That means that while the scale of the CTA's service is impressive, any delay or disruption has an outsized impact on passengers. A consistent and predictable mass transit system connecting such a massive population center is necessary to foster sustainable economic growth, improve environmental quality, and enhance social equity by providing affordable mobility.

In this project, we used machine learning to forecast ridership, identify trends, and cluster stations by usage patterns. We implemented a supervised learning workflow comparing three distinct model families: Ordinary Least-Squares (OLS) as a linear baseline, Decision Tree and Random Forest regressors to capture non-linear interactions, and Huber Regression to provide a robust estimation resilient to ridership outliers and multicollinearity. Our supervised models forecast daily ridership by station based on covariates such as day type, season, ADA compliance, lines available, and whether pre, during, or post-COVID. Our unsupervised learning analysis combined Principal Component Analysis (PCA) for dimensionality reduction of high-dimensional spatial features with K-Means clustering to identify five distinct station profiles, providing potential archetypes of stations by which to optimize.

As Chicago residents, our team was highly motivated to seek a project with the potential to improve quality of life in our own community. Providing the CTA with station-level predictive insights enables planners to move beyond broad system-wide averages to a precision-based model where they can dynamically adjust 'L' car counts and schedules to match actual demand and identify groups of stations that justify the addition of rapid transit lines Many other mass transit systems have begun utilizing machine learning and predictive modeling to optimize their own transportation. Notably, a similar project by Singapore's Land Transport Authority uses real-time data from fare cards and sensors to optimize bus routes (Singapore Management University, n.d.). Our work differs in applying supervised and unsupervised models to Chicago's rail network, focusing on predicting ridership peaks and optimizing train frequency rather than integrating multimodal transport data. Additionally, others have used Chicago transit data to build interactive visualizations (Chauhan, 2018), which is valuable, but our project extends those capabilities to predictive modeling.

Our analysis found that non-linear, tree-based predictive models effectively forecast station-level ridership despite a significant post-pandemic decline, while clustering successfully categorized Chicago's "L" stations into five distinct usage profiles based on their surrounding neighborhoods' zones and ridership. The Decision Tree model was our most effective supervised model, achieving an R^2 of 0.846, significantly outperforming our OLS baseline R^2 of 0.558. This revealed that ridership patterns are complex and non-linear relationships between geography, time, and other features that linear models struggle to capture. The most significant negative predictor was time, particularly whether pre- or post-COVID, and we clearly see a significant decrease in ridership since Chicago's initial city shutdown in March of 2020. Longitude, red line access, and whether the station was within a mile of the downtown core (zone DC) were also significant predictors. We used PCA to reduce over 100 spatial and zoning variables into four primary dimensions of variance, which we then fed into K-Means. Our K-Means clustering used these principal components to identify five unique categories of stations, including one

cluster of high-traffic, downtown core stations averaging over 7000 rides a day and one of 64 residential and local stations that provide low but reliable ridership.

Related Work

As public transit is a matter of global importance, various projects across the world have also sought to predict transit ridership for their respective transit systems. A similar project by Singapore Management University (SMU) researchers uses real-time data from Singapore Land Transport Authority fare cards and sensors to anticipate commuter demand. Their main goal was to reduce "last-mile" friction by predicting rider behavior, and they were able to predict precise bus disembarkation with 90% accuracy. Our work differs in applying supervised and unsupervised models to Chicago's rail network, focusing on predicting ridership peaks and optimizing train frequency rather than integrating multimodal transport data.

Additionally, researchers have analyzed Chicago's transit specifically. A project by data scientist Arnold Oduor (2022) uses the CTA Daily Boarding Totals dataset to forecast next-day ridership on bus and rail systems. Oduor's work focuses on comparing the performance of traditional statistical models to that of deep learning models, finding RNN models worked well. A key difference between his work and ours is that we hope to predict ridership on any given day, not just the next day, and our main interest is in how various factors such as accessibility and prevalence of work-from-home impact 'L' ridership specifically.

Cody Dirks (2022), Lead Machine Learning Scientist at data science firm Strong Analytics, built a forecasting tool for predicting daily ridership for individual 'L' lines. He did so by combining daily ridership data from 2008 to 2018 with outside information like weather, holidays, and Chicago Cubs home games, and the validation data fell within the model's 95% confidence intervals 94% of the time. This project is the most similar to ours, but they differ in that the Strong Analytics team's model predicts ridership by line, which there are only eight of, while our model predicts ridership by station, of which there are 146. This increase in granularity from line-level to station-level is critical because while line-level forecasts provide a broad operational overview, station-level predictions allow precise resource allocation and crowd management at specific bottleneck stations. Additionally, their work does not consider ADA accessibility or the impact of COVID, which we believe are essential factors.

Data Sources

The main dataset we used is "CTA - Ridership - 'L' Station Entries - Daily Totals," which is freely available in CSV format from the transit portion of Chicago's Data Portal and shows system-wide boardings across each L-station by day since 2001. Except for the NOAA data, all datasets were retrieved programmatically using Python's requests library to fetch the most current data directly from the Chicago Data Portal's download URLs. This approach ensures the workflow is reproducible and handles the large volume of records—particularly the 1.3 million ridership entries—efficiently.

CTA Daily Ridership: The foundational dataset, "CTA - Ridership - 'L' Station Entries - Daily Totals," provides longitudinal boarding data (Chicago Transit Authority, 2026).

- **Time Period:** The dataset covers the period from January 1, 2001, to October 31, 2025.
- **Record Count**: There are about 1.3 million rows of records with 5 columns of features.
- **Key Variables:**
    - *station_id:* Unique numerical identifier for each station.
    - *date:* The specific day of recorded boardings.
    - *daytype:* A categorical variable identifying the day as W (Weekday), A (Saturday), or U (Sunday/Holiday).

- *rides:* The total number of boardings per station per day.

The first few lines of the ridership dataset before cleaning or joining—

	station_id	stationname	date	daytype	rides
0	40350	UIC-Halsted	01/01/2001	U	273
1	41130	Halsted-Orange	01/01/2001	U	306
2	40760	Granville	01/01/2001	U	1,059
3	40070	Jackson/Dearborn	01/01/2001	U	649
4	40090	Damen-Brown	01/01/2001	U	411

To facilitate geospatial analysis, we joined the ridership dataset with "CTA - System Information - List of 'L' Stops," which has location information, which lines are available, the directions lines go, and accessibility information for each 'L' station. The CTA rail system map is in the Appendix for reference.

CTA System Information: The "CTA - System Information - List of 'L' Stops" dataset provides information about each place within the CTA system where a train stops (Chicago Transit Authority, 2025).

- **Time Period:** The dataset represents the current status of each station as of November 19, 2025. We were not able to find a freely available dataset with historical station records. Because of this, some station data could be incorrect time-wise, but stations are seldom built or closed and their ADA and line access very rarely changes after construction, so we do not think this is a major issue, though it is a noteworthy limitation in our research.
- **Record Count**: There are 302 rows with 16 columns.
- **Variable Significance:** This dataset provides the location (latitude and longitude coordinates), lines served (e.g., Red, Blue, Green, Brown, etc.), and ADA accessibility status for each station. This allowed for the aggregation of ridership by geographic cluster and transit corridor.

The first few lines of the system information dataset before cleaning or joining—

	STOP_ID	DIRECTION_ID	STOP_NAME	STATION_NAME	STATION_DESCRIPTIVE_NAME	MAP_ID	ADA	RED	BLUE	G	BRN	P	Y	Pnk	O	Location
0	30162	W	18th (54th/Cermak-bound)	18th	18th (Pink Line)	40830	True	False	False	False	False	False	False	True	False	(41.857908, -87.669147)
1	30161	E	18th (Loop-bound)	18th	18th (Pink Line)	40830	True	False	False	False	False	False	False	True	False	(41.857908, -87.669147)
2	30022	N	35th/Archer (Loop-bound)	35th/Archer	35th/Archer (Orange Line)	40120	True	False	False	False	False	False	False	False	True	(41.829353, -87.680622)
3	30023	S	35th/Archer (Midway-bound)	35th/Archer	35th/Archer (Orange Line)	40120	True	False	False	False	False	False	False	False	True	(41.829353, -87.680622)
4	30213	N	35-Bronzeville-IIT (Harlem-bound)	35th-Bronzeville-IIT	35th-Bronzeville-IIT (Green Line)	41120	True	False	False	True	False	False	False	False	False	(41.831677, -87.625826)

In the joined dataset (below), each row represents one day's ridership at one station overall. Each station may have multiple platforms, but each only has one overall location, ADA status, and line access information.

	station_id	stationname	date	daytype	rides	MAP_ID	STATION_NAME	latitude	longitude	ADA	RED	BLUE	G	BRN	P	Y	Pnk	O	N/S	E/W
0	40350	UIC-Halsted	2001-01-01	U	273	40350.0	UIC-Halsted	41.875474	-87.649707	1.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0
1	41130	Halsted-Orange	2001-01-01	U	306	41130.0	Halsted	41.846780	-87.648088	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	1.0	0.0
2	40760	Granville	2001-01-01	U	1059	40760.0	Granville	41.993664	-87.659202	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0
3	40070	Jackson/Dearborn	2001-01-01	U	649	40070.0	Jackson	41.878183	-87.629296	1.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0
4	40090	Damen-Brown	2001-01-01	U	411	40090.0	Damen	41.966286	-87.678639	1.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	1.0	0.0

Daily Weather Summaries: Additionally, we added weather data for each date. We accessed this data via the National Oceanic and Atmospheric Administration's climate data search tool. Due to the size of the data, we were only able to request and then download two years of daily summaries at a time, so there are

13 CSV files of weather data. We vertically concatenated them to get a continuous span of data from 2000 to 2026. (NOAA National Centers for Environmental Information, 2026)

- **Time Period:** The dataset includes daily weather summaries for all available weather stations in the city of Chicago from January 1, 2000, to February 1, 2026, which encompasses the full time frame in our ridership and system information dataset.
- **Record Count**: There are about 10 thousand rows of records with 19 columns of features.
- **Key Variables:**
    - *NAME:* This is the unique name of the weather station at which these weather metrics were collected.
    - *DATE:* The date to which the daily summary corresponds.
    - *TAVG:* Average temperature
    - *PRCP:* Inches of precipitation
    - *WT[01-22]:* Binary values that indicate the presence of various weather types, such as—
        - 01 = Fog, ice fog, or freezing fog (may include heavy fog)
        - 02 = Heavy fog or heaving freezing fog (not always distinguished from fog)
        - 03 = Thunder
        - Etc. See full documentation in the Appendix.

Here are the first few rows of the weather data after vertically concatenating but before further cleaning—

	STATION	NAME	DATE	AWND	PRCP	SNOW	SNWD	TAVG	WT01	WT02	WT03	WT04	WT05	WT06	WT07	WT08	WT09	WT10	WT11
797002	USW00014819	CHICAGO MIDWAY AIRPORT, IL US	2000-01-01	7.61	0.00	NaN	NaN	43.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
799195	USC00115110	LITTLE RED SCHOOL HOUSE, IL US	2000-01-01	NaN	0.00	0.0	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
793169	USC00124008	HOBART 2 WNW, IN US	2000-01-01	NaN	0.02	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
800657	USC00111577	CHICAGO MIDWAY AIRPORT 3 SW, IL US	2000-01-01	NaN	0.00	0.0	0.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
792438	USW00094846	CHICAGO OHARE INTERNATIONAL AIRPORT, IL US	2000-01-01	7.38	0.00	0.0	0.0	42.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

We joined the weather data with the main ridership dataset on the date feature. The resulting dataframe is just like the dataset did before, just with those additional weather columns.

Chicago Zoning Districts: To further enrich the dataset, we decided to overlay Chicago zoning data to better understand the sort of area (i.e. residential, commercial, etc.) each station exists within. We accessed this dataset, "Boundaries - Zoning Districts (current) - Map," in CSV format from the Chicago Data Portal. It shows the district land use boundaries for each unique district. (City of Chicago, 2026)

- **Time Period:** The dataset reflects current zoning as of February 18, 2026. Like stations, though, zones do not often significantly change, and in the Chicago Data Portal, it says this dataset was first created on January 4, 2016, and “No changes have been archived yet.” While that may mean that no older versions were archived rather than that no older versions (with zoning changes) exist, it is the best data we are able to access freely at this time.
- **Record Count**: There are about 14.9 thousand rows of records with 25 columns of features.
- **Variables of Interest:**
    - *the_geom:* This is a multipolygon of latitude and longitude coordinates that show the exact boundaries of each unique zoning district. These were used to calculate the proximity of each 'L' station to specific zoning types.
    - *ZONE_CLASS:* This is a value like “B-3” or “R-1.” The letter or letters before the dash indicate the zone type, such as business or residential, and the number after the dash represents how many buildings can be built on a lot within that zone, according to the Municipal Code of Chicago.

Feature Engineering

We began by ensuring data integrity and type standardization before joining any datasets so that we could cleanly tell if any issues came up from improperly joining rather than messy data to begin with. This included converting date strings into datetime objects and categorical variables (like “daytype” and “DIRECTION_ID”) into the appropriate category types. We also processed the rides column by casting the values to integers to allow for mathematical analysis. Next, we parsed the "Location" string column to extract numerical latitude and longitude values by stripping parentheses and splitting the strings. These corrections ensured we would not face computational errors due to data integrity later in the pipeline.

To prepare our weather data, we vertically concatenated the 13 individual weather CSV files. Because on some days, multiple weather stations in Chicago recorded data for the same date, we prioritized data from the O’Hare International Airport station because it was the only station with data available for every date, and it was important to keep consistent as weather can vary by station. We also cleaned the weather type binary columns by filling missing values with zeros and renaming generic codes, such as WT01, to their human-readable definitions, like “Fog_Ice_Fog” to improve interpretability. Once the data was fully cleaned and deduplicated, we performed a left join with the ridership data on the date column. This ensured that every daily ridership record was augmented with weather information such as average temperature and inches of precipitation.

Then, we aggregated the system information dataset to prepare for joining. To ensure no issues, we checked for stations with missing values or conflicting ADA or latitude and longitude information ahead of this, but we found no issues. We grouped the transit system information by “MAP_ID” to ensure each station had one unique entry, preventing duplicate rows during the join with the ridership time-series data. In aggregating, we used the max() function for the line colors so that for each station, a line variable is equal to 1 if that line is available at that station. We also constrained the dataset to only the relevant columns for later analysis. Sometimes the same station has multiple names, such as Western-Cermak, which is also called and sometimes referred to officially as Western. Because of discrepancies in station names, we could not join the datasets by name. Additionally, many stations (such as 18th, which appears in the second chart in the Data Sources section) have more than one place where trains stop, such as a platform for trains heading west and a platform for trains heading east. Because of this, a single station may have multiple “STOP_ID” values, but each only has one unique “MAP_ID”. As the daily ridership dataset only tells us information at the station level, not the stop level, we performed a left join to combine the daily ridership entries with the system information dataset using the station identifier (“station_id” and “MAP_ID”). The resulting dataframe is clean and represents one row per station and date, which was our desired structure for later modeling.

To integrate Chicago’s zoning districts, we used GeoPandas to convert the zone boundaries into geometric objects. We projected the station coordinates and zoning boundaries onto a coordinate system then generated 0.5-mile and 1-mile radiuses around each station’s geographic coordinates. To address data noise, we performed coordinate verification to ensure all stations were accurately mapped before geospatial processing. We then performed geospatial intersection to create binary indicators such as “residential_multi_unit_district_within_mile” and “heavy_industry_district_within_half_mile” that identify whether each zone is present within each station’s 0.5-mile and 1-mile radiuses, respectively. See the map in the Appendix for how this was plotted. We then left joined these zone indicators onto the main dataset on the “MAP_ID” feature.

Once we had this one dataframe of ridership data, station and weather information, and zoning, we moved from cleaning to feature engineering. Our goal was to transform existing features such that they would be most useful for predictive modeling. We created several new variables from the date column to capture temporal trends: “days_since_2000” is a continuous count of days since January 1, 2000, to serve as a

numerical trend variable; month and year are individual integer features representing each date's respective month and year; and season breaks up months into "Winter," Spring," "Summer," and "Fall" as we suspect the influx of tourists in the summer may have an impact on ridership. We also introduced the binary feature "post_shutdown", which represents whether the date was before or after Chicago's March 13, 2020, shutdown of schools due to COVID-19. This allows the model to account for the significant and lasting shift in transit usage following the onset of the pandemic.

We also transformed the various train directions into binary features where N/S represents Northbound or Southbound service and Eastbound or Westbound service. We needed to include both of these rather than dropping one because stations can have just North/South service, just East/West service, or both North/South and East/West service. Again, we did not drop any features because it is possible for a station to access multiple lines. We also convert the daytype categorical variable into one-hot encodings to distinguish between weekdays, Saturdays, and Sundays/Holidays. Beyond transit-specific features, we incorporated the zoning proximity flags and daily weather metrics. By encoding these, we are providing the model with the mathematical signals it needs to identify seasonal and weather trends as well as service patterns. To rigorously evaluate our models, we used a chronological (or time-series) train-test split based on the "days_since_2000" feature. We used a cutoff of 25 years (approximately 9131 days), designating all data through December 31, 2023, as our training set and all data from January 1, 2024, onwards as our test set. This time-based approach ensures that our performance metrics—such as the decision tree's 0.85 R^2—reflect the model's ability to forecast future transit demand based on historical patterns, which is more representative of real-world transit planning than a random split. This is essential as forecasting future data is much harder than just understanding the patterns of past data.

See the Appendix for a list of final features and their data types.

Part A. Supervised Learning

Methods

Our supervised learning workflow targets "rides" as a continuous numerical variable. To validate our model, we used a time-series split to respect the chronological nature of ridership data and prevent data leakage. We evaluated three distinct types of models to capture the varying complexities of transit patterns.

- **Ordinary Least Squares (OLS) Regression:** We chose OLS as our baseline model because our target variable is numeric and continuous and we have a very large data set. Another benefit is that OLS is highly interpretable and quick to train. If OLS performed well, it would suggest the relationship between weather and location and ridership was additive and constant—a simpler, more maintainable solution. Using OLS first also helped identify if features like "days_since_2000" and "season" were statistically significant before we hid them inside a "black box" model. While OLS provided useful signals about particular variables of interest, ultimately the performance was not satisfactory with an R^2 of 0.5581 on the test data.
- **Huber Regression:** In analyzing ridership over time, we noticed many outlier events, such as St. Patrick's Day and Lollapalooza, had extremely high ridership. As it was not feasible for us to add in every major event in Chicago since 2000 as a feature, we selected a Huber model for robustness. Unlike OLS models, which optimize based on squared error loss, the Huber Regressor uses a quadratic penalty for small errors but switches to a linear penalty for outliers, so it is more "stubborn" to unusual ridership, such as during major holidays and events. We chose Huber over something like Lasso Regression because we found that many features mattered in predicting ridership. Our results were slightly improved from OLS, with an R^2 of 0.6246 on the test data, but

this score signaled to us that the data's underlying structure was likely more jagged or hierarchical.

- **Decision Tree Regression:** To address the potentially jagged or hierarchical structure of the underlying patterns, we moved to a tree-based model. A decision tree made sense because transit data can be inherently complex as ridership does not increase linearly with features like temperature, for example, but plateaus at comfortable thresholds and decreases but never hits zero during extreme cold. Decision trees naturally capture non-linearities and complex feature interactions without requiring manual polynomial transformations. Using a decision tree model significantly improved performance, resulting in a 0.8464 R^2 score on the test data. Because this performance was strong, we also tried a random forest model, which also has a tree-based architecture, but it did not perform quite as well.

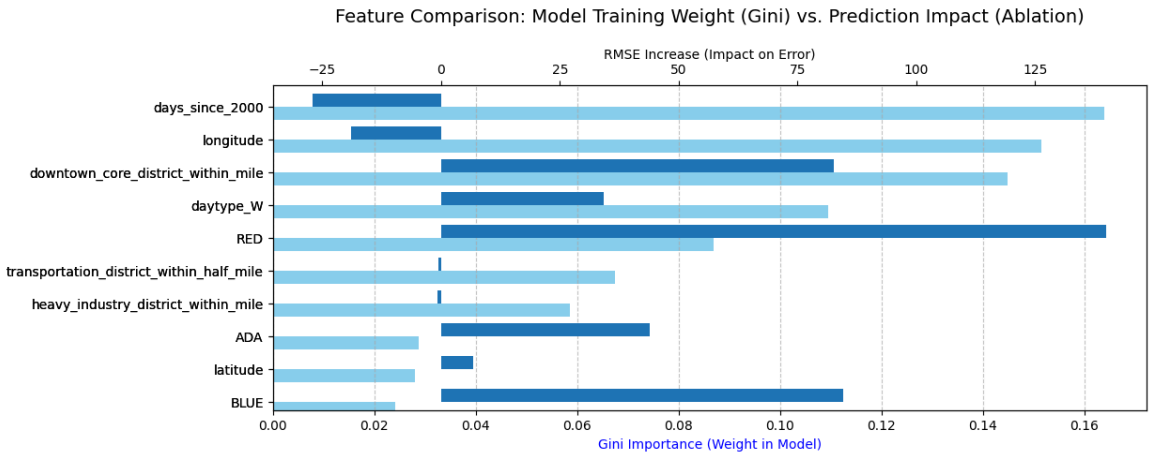
Evaluation

Overall, we selected Root Mean Squared Error (RMSE) and R^2 as our primary evaluation metrics. R^2 provides an intuitive measure of how much variance in ridership our features explain, while RMSE provides a real-world error value in terms of actual passenger counts, making the error easier to communicate to transit planners. To ensure the reliability of our results beyond a single split, we used 5-fold time-series cross-validation. As previously discussed, this method respects the temporal order of the data by chopping it into folds. For example, one fold may train on 2000-2005 data then validate on 2006 data. The decision tree model outperformed the other families significantly in terms of R^2 and RMSE and showed the lowest standard deviation, indicating that it is the most stable model across different time periods. The high standard deviations for the OLS and Huber models may reflect the sudden, unexpected drop in ridership due to COVID.

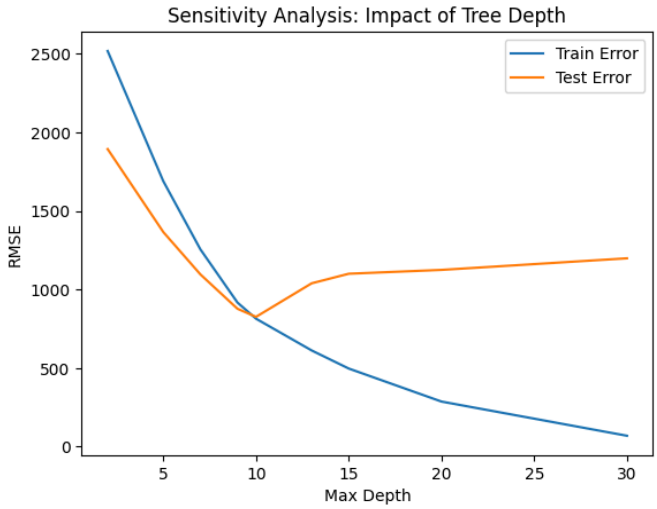
Time-Series Cross Validation Results

Model Family	Best Model	Mean R^2	Std. Dev. R^2	Mean RMSE
Linear	OLS Regression	0.1447	± 0.7304	2,112.22
Robust Linear	Huber Regressor	0.0810	± 0.6694	2,284.20
Tree-Based	**Decision Tree**	**0.7679**	**± 0.2558**	**1,032.80**

Once we selected the decision tree as our best model, we analyzed feature importance. *See the [full list of decision boundaries](#) in the Appendix.* Gini importance showed that time, geographic location, and line availability were the most important. To verify this, we performed an ablation study, removing the top ten features one-by-one to measure the increase in our model's RMSE.



As is evident in the chart above, “RED” (whether the station is on the red line) is the most critical feature. Removing it caused the largest spike in RMSE. The next most important feature was “downtown_core_district_within_mile,” revealing that proximity to the Loop was vital. We also performed sensitivity analysis by testing the sensitivity of our results to the decision tree model’s max_depth hyperparameter. We found that the model is highly sensitive to depth and underfits below 10 and overfits above it.



We identified various tradeoffs between the models. The main two were interpretability versus accuracy and training speed versus resilience. In reference to the former, while OLS allowed us to see exactly how many rides each degree of temperature added, making it highly interpretable, it was about 30% less accurate than the decision tree model. We traded clean and simple interpretability for accurate predictions. In regard to the latter, the decision tree is significantly faster to train than iterative models like the Huber Regressor, but it lacks smoothness near decision boundaries. Due to the hierarchical nature, the decision tree model’s predictions may jump abruptly.

Moving on to failure analysis, we identified the three instances (date x station) in which the decision tree model’s predictions were farthest from accurate.

Date	**Coordinates (Lat, Long)**	**Day Type**	**Actual Rides**	**Predicted Rides**	**Absolute Error**
March 15, 2025	41.891, -87.628 (Grand)	Weekend	24,425	6,940.51	**17,484.49**
March 15, 2025	41.885, -87.630 (Clark/Lake)	Weekend	25,011	8,116.13	**16,894.87**
Oct 12, 2025	41.853, -87.631 (Cermak)	Weekend	15,472	1,891.17	**13,580.83**

March 15, 2025, was St. Patrick’s Day, where Chicago dyes the river green. This popular event concentrates the local and tourist population in a very small part of the city with only Grand and Clark/Lake being reasonable stops at which to exit for viewing. October 12, 2025, was the Chicago Marathon, and the Cermak-Chinatown stop is where a lot of people go to watch runners as it has a large cheering section designated by the race and is part of the last few miles of the marathon. The fourth most inaccurate prediction instance was during Pride weekend last summer. This leads us to conclude that holidays and major events in the city are associated with outsized ridership and are what our model has

the hardest time predicting. Future models should take major holidays into account, particularly at stations near where events are scheduled to take place.

The second category of failure we noticed was due to severe weather. For example, there were deep freezes on January 5, 19, and 20 of 2025 that resulted in lower ridership than predicted across stations. Severe weather is particularly important to ridership at the O'Hare station because flights are canceled when there are severe weather events. Ridership at that station follows flight schedules much more than it follows typical ridership rules, making it difficult for our model to predict. To better capture these impacts, the model would benefit from a categorical "weather alert" feature in addition to a simple numerical temperature. Ridership behavior often follows a non-linear threshold—once a weather event is deemed severe by the city, the decision to stay home becomes a binary choice that a linear variable fails to capture. At O'Hare, a binary "ground stop" feature may better capture the sudden decrease in airport transit demand during storms.

The final category of failure present in our data occurred when stations were closed for part but not all of a day for construction. When stations fully close, this is evident, but partial closes are hard to predict. For example, there was a particularly inaccurate prediction on October 25, 2025, at State/Lake as it underwent construction. Ideally, adding a service status feature that identifies whether stations are fully or only partially open would likely improve performance. Given that construction is scheduled in advance by the CTA itself, however, they likely already have an understanding that ridership will decrease regardless of a model's predictions.

Part B. Unsupervised Learning

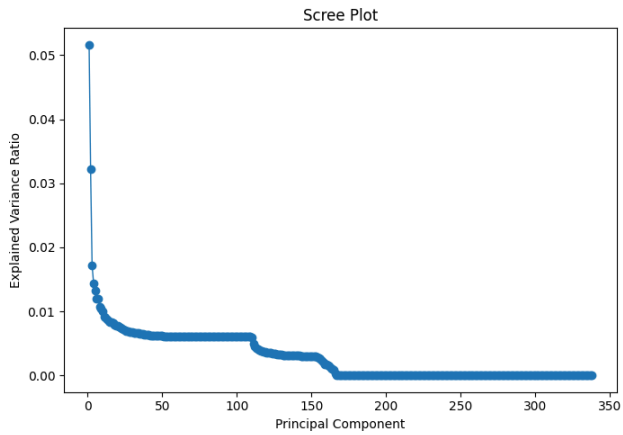
Methods

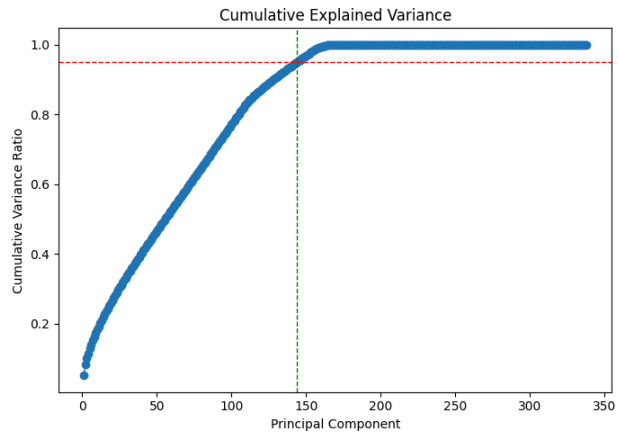
The main goal of our unsupervised learning workflow was to identify latent station profiles by analyzing the relationship between ridership volume and geographic and zoning features. We used a two-stage pipeline: PCA for dimensionality reduction followed by K-Means Clustering to categorize stations. The feature representations focused on high-dimensional spatial, zoning, and ridership data, including over 100 variables describing the proximity of various land-use districts within a 0.5-mile and 1-mile radius of each station. Our project considers physical locations that have inherent grouping, for example the stations on the Red Line already form a group due to their existing connection. Thus, this analysis should provide additional groupings that go beyond what is already connected and may help suggest improvements.

- **Principal Component Analysis (PCA):** We chose PCA to reduce the complexity of our many variables into four primary dimensions of variance. PCA is a probabilistic method that can help prevent the curse of dimensionality from skewing our results. In particular, it can help avoid overfitting. By distilling the most significant variance into a few components, we ensured that downstream clustering was based on meaningful patterns rather than in redundant or sparse data.
- **K-Means Clustering:** Following PCA, we applied K-Means, which is an instance-based, non-probabilistic clustering algorithm, to identify five distinct station profiles. We selected this method because it is efficient and able to produce clear, distinct clusters—such as high-traffic downtown hubs or low-volume residential stops. This allowed us to move beyond obvious groupings such as existing station line color to find hidden patterns in the data.

We performed hyperparameter tuning for both models to ensure stable results. For PCA, we evaluated the explained variance ratio using a Scree Plot (below) to identify the optimal number of components to capture most of the signal before capturing noise. We discovered that we had 3 major principal components to consider with PCA. We also mapped the cumulative explained variance, which shows that

at 144 principal components, we achieve 95% variance. This performance is why we will be focusing only on the top 3 components moving forward. We included the breakdown in separate tables of the component pieces used for PC 1, 2, and 3 here. This allows us to understand which features led to the most variability. Concentrating on these will allow us to improve our supervised model by decreasing the size of the model but maintaining accuracy. Each principal component is composed of different features each contributing and influencing it to differing degrees. Understanding which features contribute to the most impactful principal component can lead to insights on important features to our model.





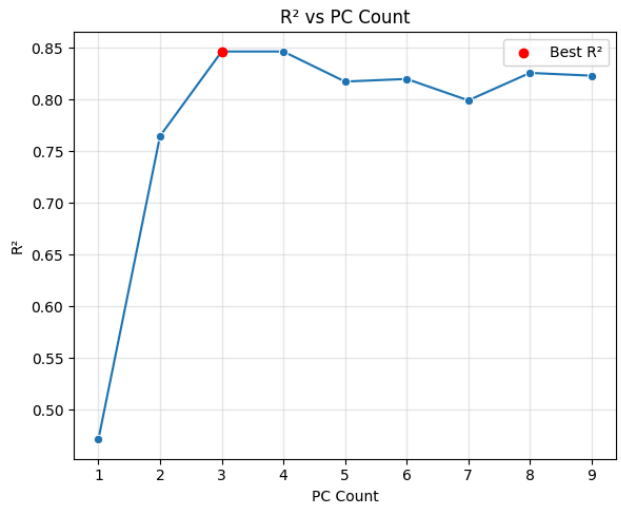
Evaluation

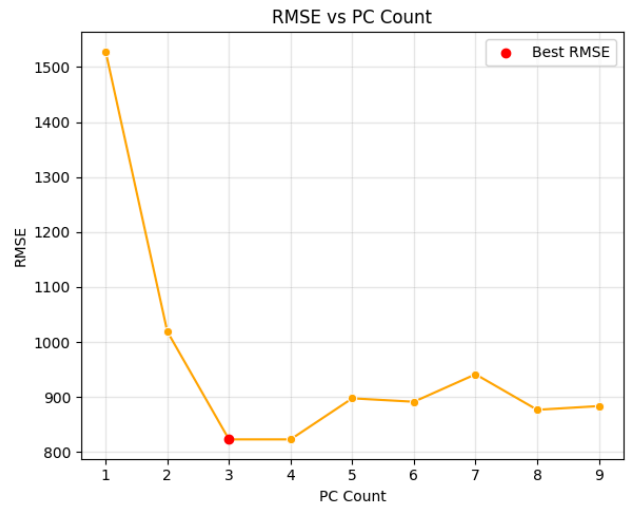
We chose to evaluate PCA with an explained variance ratio because it quantifies how much information was retained after compression, which allowed us to identify the point of diminishing returns. We evaluated our K-Means clusters with a silhouette score, which allowed us to measure cluster cohesion and separation to identify how similar stations in each cluster are to each other and how distinct they are from other stations. Finally, we again used R^2 as an evaluation metric to quantify the real-world effectiveness of our principal components and clusters when used in a decision tree model for prediction. Evaluating the clustering model with a silhouette score and ablation testing revealed that the clusters, while distinct, are quite weak. The base silhouette score for our model was **0.2426** from a range of -1 to 1, which is a low positive score indicating weak clustering. This is still better than a negative score, which means that the center of a cluster is, on average, closer to points belonging to another cluster, which means there was incorrect clustering. While there are observable clusters from a geography and ridership standpoint, ridership across the CTA system is quite continuous. This means that stations across each cluster are more similar than one might think. Therefore, every station sees some amount of consistent use and should be viewed as important, even as there are disparities between traffic and location.

To gain a real world understanding of what each principal component represents, we looked at the most important features for each. The full breakdown is in the Appendix. Principal component 1's most contributing features include half mile proximity to community shopping districts, two flat townhouse multi-unit districts, and neighborhood shopping districts. Principal component 2's most important features include proximity to downtown mixed use districts, downtown residential districts, and downtown service districts. Finally, principal component 3 seems to be more influenced by line information like whether trains travel east and west through the station or north and south, whether the ride happened on a red line, and proximity to a transportation district.

Once we identified the patterns using PCA, we then created a new supervised model to see if we can maintain accuracy compared to our original decision tree model while using fewer features. The updated model is trained only on our principal components. Our left graph (below) demonstrates that our accuracy

peaks at 3 principal components with an R^2 of 0.8466. We also see that including 4 or more components starts to decrease and plateau our accuracy with our new model. Our RMSE score shows our error steeply decreases as we reach 3 principal components, after which it rises at 4 and beyond. This means our new model with 3 principal components is performing similarly to our original model which used many more features.





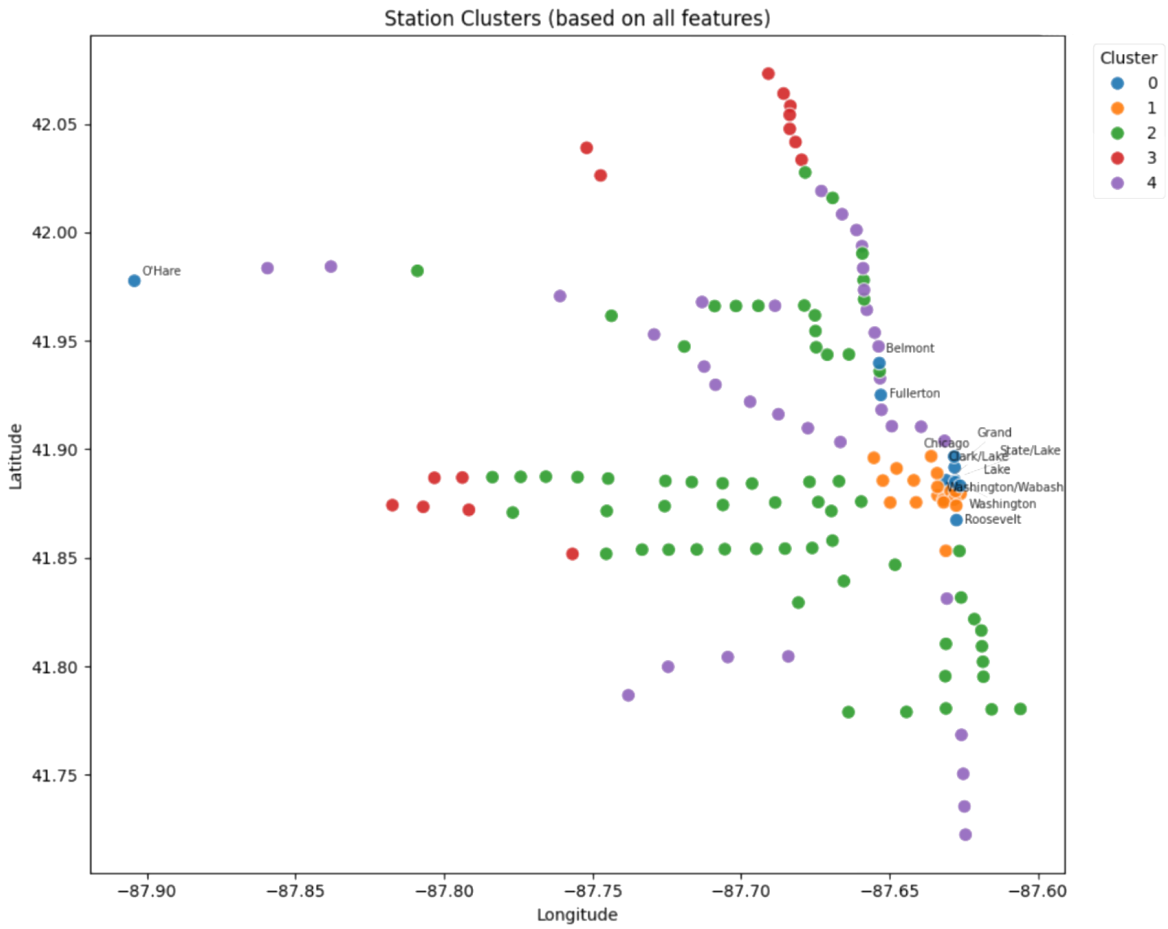
Our second model uses K-Means Clustering. This method groups similar points of data together into similar clusters. For our model, we used station average ridership numbers based on year and season as features so we could group stations with similar ridership numbers and other features in our station information together. When we map those clusters onto stations, you can see that individual 'L' lines do not all appear the same and that ridership varies across the city. While downtown contains the highest ridership it also contains some stations in the lowest ridership within close proximity.

```
Average ridership statistics by cluster:
```

cluster	n_stations	mean	median	std	min	max
0	11	7179.05	6664.45	1286.57	5733.98	10174.25
1	20	2757.65	2852.42	890.06	1339.25	4172.97
2	63	957.68	904.13	405.76	257.59	1878.81
3	15	833.85	591.93	514.96	423.55	1926.38
4	35	2897.95	2750.68	872.35	1917.27	5050.67

When looking at the clusters (on the next page), we can see that the clusters are not all uniform in size but that some clusters are much larger than others. Cluster 0 seems to be high traffic. centered around the Loop, O'hare, Fullerton (a commonly used transfer stop between the Red and Brown lines) and Belmont (close to Wrigleyville and the famous Wrigley Stadium, home of the Chicago Cubs baseball team). Cluster 1 and 4 are medium traffic stations with stations in Cluster 1 being closer to the Loop on average than stations in cluster 4. Clusters 2 and 3 are lower traffic stations with stations in Cluster 2 on average being closer to the Loop than those in Cluster 3. You can see this through the table and mapping provided as the latitude and longitude track to the city architecture. We only labeled cluster 0 in this map to

highlight the stations with the highest average traffic. While the labels downtown overlap, this confirmed our understanding that O'Hare and the Loop are a part of the stations with the highest traffic.

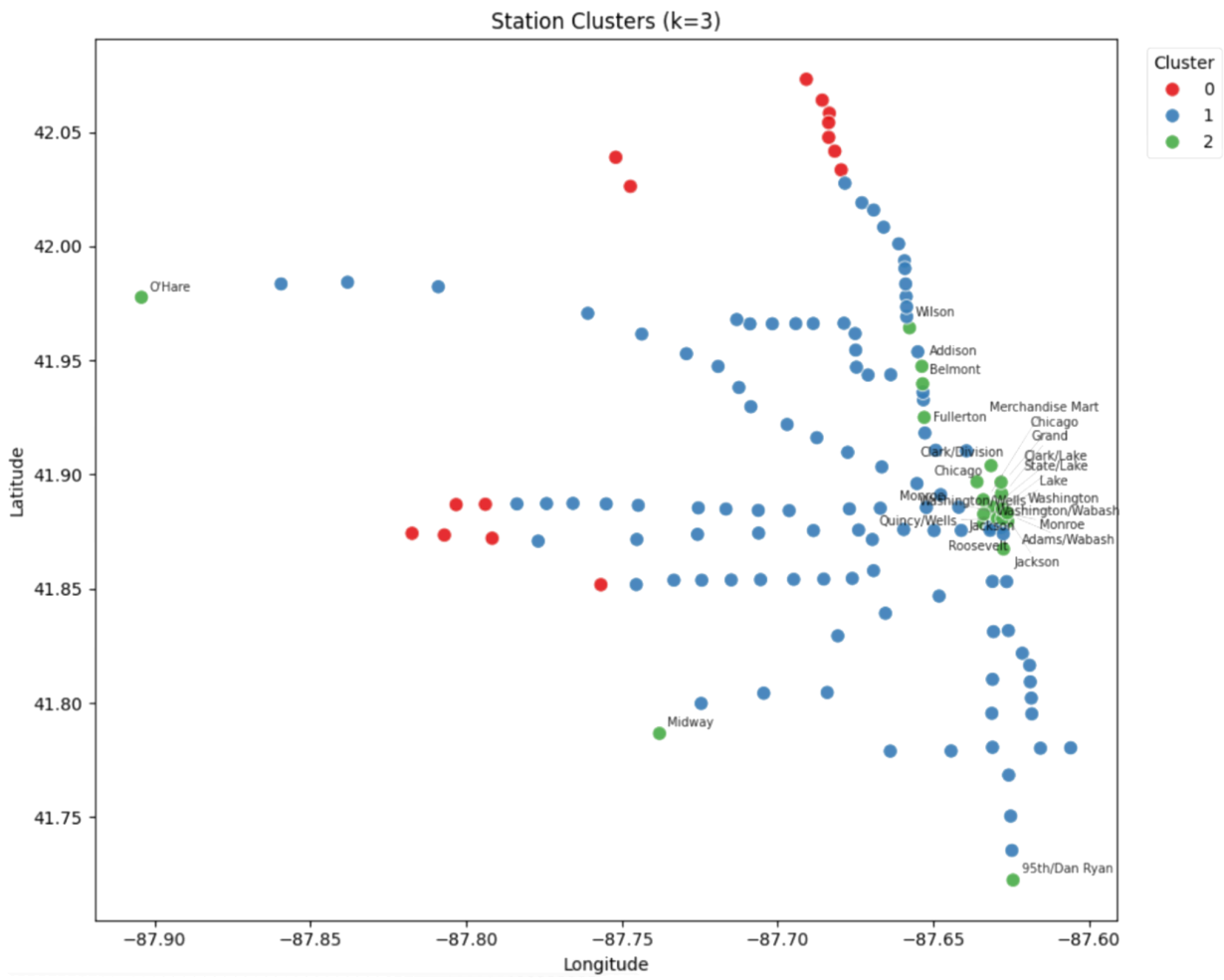


We can get stronger clustering by limiting the number of clusters. Below is a table of our ridership and station data using k=3 clustering, giving us fewer but more distinct clusters. Using fewer clusters got us a better silhouette score of **0.3169**, which is an improved, but still weak score. This score came at the cost of nuance as we lost details on stations' proximity to the loop. (See map on the next page.) Furthermore, the average ridership for the high traffic stations cluster plummeted as we went down to 3 clusters. The high traffic cluster for K=5, Cluster 0, had an average ridership of 7179.05. The high traffic cluster for K=3, Cluster 2, had an average ridership of 5377.97. The lowest traffic clusters of both K=5 and K=3 are both 833.85. While the clustering score improved, what separates a higher traffic cluster from a lower one decreased as we went from 5 clusters to 3. In this visualization, cluster 2 is the high traffic cluster which we labeled. While the Loop and O'Hare remain in the high traffic cluster, we saw a few additions to include Midway, which is also an airport, and a few other stations.

cluster	n_stations	mean	median	std	min	max
0	15	833.85	591.93	514.96	423.55	1926.38
1	104	1552.26	1445.91	882.29	257.59	3754.76
2	25	5377.97	4791.97	1877.35	3273.67	10174.25

Adding cluster labels to the supervised learning model had varying effects on performance. The observed RMSE from a model with k=5 cluster labels added to the data is 815.14. The observed R^2 for the K=5 cluster-labeled model is 0.8210. Inserting the k=3 labels gave us an R^2 of 0.8470 and RMSE of 824.38. Given this, the K=3 clustering-labeled model is the best model for improving our ability to predict ridership.

When we observe this, we can see visually that there are some high traffic zones that may benefit from more service coverage or potentially better connectivity. We do not suggest eliminating zones with lower ridership clusters but suggest that there is potential to add additional bypass or direct connections to some of these high ridership zones. One might improve service between high traffic zones by connecting the O'Hare station directly to downtown with an express or separate service line.



Discussion

In building our supervised learning model, we learned that geography (latitude, longitude, and zoning) consistently outperforms temporal features and is highly important. We also learned that the relationship between our many features and ridership is non-linear and that tree-based models perform well. Additionally, we saw that while there are over 100 features, downtown proximity, Red line access, and date do most of the heavy lifting. We were surprised that the Huber Regression, with its robustness to outliers, did not outperform OLS by very much. Perhaps outlier events like the Chicago Marathon or St.

Patrick's Day are too extreme even for a robust estimator. Finally, we found it interesting that even in recent data, post-COVID is an important feature and ridership patterns have not returned to pre-2020 logic.

Our main challenges were in learning how to map the geospatial zoning data onto our stations to determine proximity and figuring out how to address the many unique stations as categories. Geopandas proved to be very helpful, as was our mentor. In terms of the station names, we decided to drop them and rely on latitude, longitude, and other system details instead to force the model to learn why certain stations have high or low ridership rather than just memorizing their names. We hope this will make the model more robust to new, added stations as well. Given more time and resources, we would like to include major city events, sports games, concerts, and conventions for better predictive power. Adding lags and rolling averages would also likely be useful and worth testing.

While we were not able to improve model performance with unsupervised learning methods, we were still able to find meaningful patterns and better understand our data. PCA allowed us to group certain features into three principal components which drove most of the variability in our data. We saw that principal component 1 contained features that related to shopping, mixed use, and residential districts. Principal component 2 contained features that related to downtown districts. Principal component 3 contained more station information like whether trains traveled East/West or North/South and what lines connected through each station. Using those principal components alone, we were able to construct a model that performed on par with our best supervised learning method, which was a surprise. Getting results on par with a 100+ feature model using just 3 principal components, suggesting that Chicago's transit-oriented development is highly structured.

Our clustering methods produced the best performing model. This came at a cost as the clustering model took significantly more resources to construct the model. PCA was able to produce similar results in a shorter amount of time. Ultimately, both of these unsupervised models produced an R^2 within .04, demonstrating that both of these tuning methods are effective. For future review, PCA would be the recommended method for ease of training. Our biggest challenge was the initial low Silhouette score, but we addressed it by doing sensitivity analysis to optimize the hyperparameter k, which resulted in reducing from 5 clusters to 3. We would extend the model by including average hourly ridership rather than just total ridership averages to better distinguish the unique patterns between commuter hubs and nightlife locations.

Ethical Considerations

Chicago remains a historically redlined city with segregation that is still inherently built into the neighborhoods, zoning, and allocation of resources. (Rothstein, 2017) With large wealth disparities and systemic discrimination baked into the structure of the city, our research is intended to serve as a tool for adding and optimizing 'L' service, not removing it. Our primary concern is allocative harm: the risk that our model's predictions will lead to a system that favors some groups while neglecting others. If the model predicts lower ridership for certain stations or certain clusters have lower ridership, and the CTA uses this to reduce service, it could create a transit "death spiral." In this scenario, lower service leads to lower ridership, which the model then predicts as a further decline. This could result in lowering mobility for minority communities that already have less access to these services in a highly segregated city. To mitigate this, we propose that our research is used to identify where service should be increased, not where it can be cut.

Secondly, we are concerned that our inclusion of "post_shutdown" as a covariate may inadvertently penalize neighborhoods with more essential workers. For example, "post_shutdown" suggests that

ridership is lower across all stations since the onset of COVID in 2020, but ridership may be down less in neighborhoods where the majority of people still work in person. As the ability to find remote work is associated with higher income households and non-minority populations, predicting lower ridership across stations regardless of the percent who work from home may disproportionately lower service to lower income and minority neighborhoods by underpredicting ridership. Similarly, features like "downtown_core_district_within_mile" reflect which parts of the city have been invested in historically. Particularly in our unsupervised model, the highest traffic cluster closely reflects the downtown core. Relying too heavily on these clusters to schedule cars could reinforce the presence of transit deserts where people are unable to access public transit.

Based on Singapore's example, we suggest that there should be a reasonable minimum of service available to all parts of the city, regardless of ridership. Public transit is paid for by all Chicagoans and should remain a public right for all Chicagoans. Insights from our clustering analysis can be used to redistribute cars to alleviate congestion, not to siphon resources from residential stations.

Statement of Work

We distributed project responsibilities dynamically to fit our existing strengths and learning goals. Frankie began the project as project manager and sourced the data, cleaned it, and joined the ridership, system information, and weather datasets. Partway through, Frankie transitioned to working more on the code and writing, performing the exploratory data analysis and writing the data sources, feature engineering, supervised learning, ethical considerations, and technical appendix sections. Maria took over the project management responsibilities in the second half, ensuring we stayed on schedule, completing the second standup, and coordinating the final narrative. She identified relevant studies and work and wrote the introduction, related work, and unsupervised and discussion sections. Alex led the geospatial and modeling portion of the work, which included cleaning and engineering our zoning dataset and developing the pipeline. He implemented the initial OLS regression and executed all unsupervised learning models. He also created and finalized our GitHub repository and map visualizations. All of us collaborated on the initial proposal and editing of the final report to ensure consistency.

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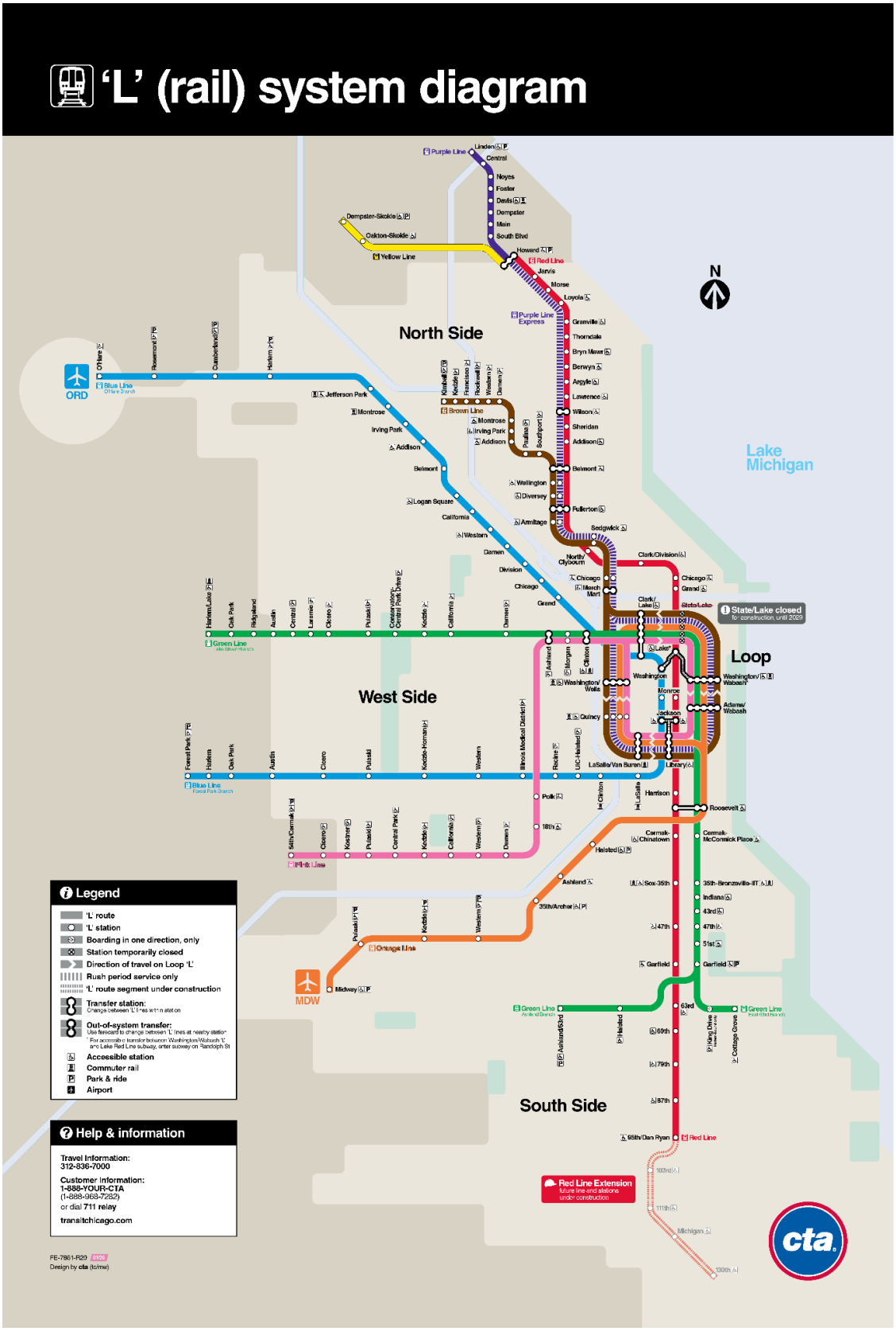
Maria Benjamin, Alexander Cao, and Frankie Pike
SIADS 696: Milestone II

Appendix

Map of 'L' Stations and Lines (January 2026)



Below is the exact text of: **GHCN (Global Historical Climatology Network) - Daily Documentation**

NOTE: English units are displayed on pdf output format; Either Metric or standard English units on csv or txt output formats depending on user specification.

I. Description

GHCN (Global Historical Climatology Network)-Daily is a database that addresses the critical need for historical daily temperature, precipitation, and snow records over global land areas. GHCN-Daily is a composite of climate records from numerous sources that were merged and then subjected to a suite of quality assurance reviews. The archive includes over 40 meteorological elements (see Table 4 below for complete list) including temperature daily maximum/minimum, temperature at observation time, precipitation, snowfall, snow depth, evaporation, wind movement, wind maximums, soil temperature, cloudiness, and more.

GHCN-Daily will serve as a replacement product for older NCDC-maintained data sets that are designated for daily temporal resolution (i.e. DSI 3200, DSI 3201, DSI 3202, DSI 3205, DSI 3206, DSI 3208, DSI 3210, etc.). It will function as the official archive for daily data from the Global Climate Observing System (GCOS) Surface Network (GSN) and is particularly well suited for monitoring and assessment activities related to the frequency and magnitude of extremes. Containing observations of one or more of the above elements at more than 100,000 stations that are distributed across all continents, the dataset is the world's largest collection of daily climatological data. The total of 1.4 billion data values includes 250 million values each for maximum and minimum temperatures, 500 million precipitation totals, and 200 million observations each for snowfall and snow depth. Station records, some of which extend back to the 19th century, are updated daily where possible and are usually available one to two days after the date and time of the observation.

Some of the data provided here are based on data exchanged under the World Meteorological Organization (WMO) World Weather Watch Program according to WMO Resolution 40 (Cg-XII). This allows WMO member countries to place restrictions on the use or re-export of their data for commercial purposes outside of the receiving country. Those countries' data summaries and products which are available here are intended for free and unrestricted use in research, education, and other non-commercial activities. For non-U.S. locations data, the data or any derived product shall not be provided to other users or be used for the re-export of commercial services.

II. Format/Observation Definitions

(note: the term ‘element ’or ‘value’ is used throughout this documentation and refers to an individual meteorological/climatological measurement or statistical value such as temperature, precipitation (amount), etc.)

Users are given the choice between the following two delivery formats:

1. GHCN-Daily Form- Portable Document Format (PDF) output giving 5 core values (see Table 4) and, if available, the following additional values: TOBS (temperature at the time of observation), EVAP (evaporation of water from evaporation pan), WDMV (24-hour wind movement), SN*# (minimum soil temperature) and SX*# (maximum soil temperature). More details about these values are in Table 4 (below). There are no flags (attributes) given with the GHCN- Daily Form pdf file other than measurement flag “T” (trace value for precipitation, snowfall or snow depth, as per table 3 below). Units for data values can be in metric or standard depending on user preference for both CSV and text output. Empty, or blank, cells indicate that a data observation was not reported.
2. Custom GHCN-Daily CSV- Output files contain .csv extension and optimized for spreadsheet

usage (i.e. delimited file). Besides unit preference, user is also given the choice whether to include flags, station name or geographic location in data request. The user can define which of the elements listed in Table 4 (below) to include in the data request.

3. Custom GHCN-Daily ASCII Form-Output is ASCII text file and the user is given the choice whether to include flags, station name or geographic location in data request. The user can define which of the elements listed in Table 4 (below) to include in the data request.

A. Data observations (values)
Each record represents all selected observations (values) available for a given station-day. The initial section of each record is ordered as follows with the following definitions:

STATION (17 characters) is the station identification code. Please see http://www1.ncdc.noaa.gov/pub/data/ghcn/daily/ghcnd-stations.txt for a complete list of stations and their metadata.

STATION_NAME (max 50 characters) is the name of the station (usually city/airport name). Optional output field.

GEOGRAPHIC_LOCATION (31 characters) is the latitude (decimated degrees w/northern hemisphere values > 0, southern hemisphere values < 0), longitude (decimated degrees w/western hemisphere values < 0, eastern hemisphere values > 0) and elevation above mean sea level (tenths of meters). This is an optional output field.

DATE is the year of the record (4 digits) followed by month (2 digits) and day (2 digits).

B. Observations (values) and flags (attributes)
Following this initial section of the record, all selected observations and flags are given in the following order:

Observation(s) | Measurement Flag | Quality Flag | Source Flag | Time of Observation | repeat for next element (when more than one element is selected), where:

Observation(s) is/are defined in Table 4 below. 9's in a field (e.g.9999) indicate missing data or data that has not been received.

Measurement Flag (attribute) is defined in Table 1 below

Quality Flag (attribute) is defined in Table 2 below

Source Flag (attribute) is defined in Table 3 below

Time of Observation is the (2 digit hour, 2 digit minute) 24 hour clock time of the observation given as the local time at the station of record.

Note: The 4 flags listed above are optional on the Custom GHCN-Daily ASCII Form.

Table 1 (Measurement Flag/Attribute)
Blank = no measurement information applicable

- A = value in precipitation or snow is a multi-day total, accumulated since last measurement (used on Daily Form pdf file)

- B = precipitation total formed from two twelve-hour totals
- D = precipitation total formed from four six-hour totals
- H = represents highest or lowest hourly temperature (TMAX or TMIN) or average of hourly values (TAVG)
- K = converted from knots
- L = temperature appears to be lagged with respect to reported hour of observation
- O = converted from oktas
- P = identified as "missing presumed zero" in DSI 3200 and 3206
- T = trace of precipitation, snowfall, or snow depth
- W = converted from 16-point WBAN code (for wind direction)

Table 2 (Quality Flag/Attribute)
Blank = did not fail any quality assurance check

- D = failed duplicate check
- G = failed gap check
- I = failed internal consistency check
- K = failed streak/frequent-value check
- L = failed check on length of multiday period
- M = failed mega-consistency check
- N = failed naught check
- O = failed climatological outlier check
- R = failed lagged range check
- S = failed spatial consistency check
- T = failed temporal consistency check
- W = temperature too warm for snow
- X = failed bounds check
- Z = flagged as a result of an official Datzilla investigation

Table 3 (Source Flag/Attribute)
Blank = No source (i.e., data value missing)

- 0 = U.S. Cooperative Summary of the Day (NCDC DSI-3200)
- 6 = CDMP Cooperative Summary of the Day (NCDC DSI-3206)
- 7 = U.S. Cooperative Summary of the Day -- Transmitted via WxCoder3 (NCDC DSI-3207)
- A = U.S. Automated Surface Observing System (ASOS) real-time data (since January 1, 2006)
- a = Australian data from the Australian Bureau of Meteorology
- B = U.S. ASOS data for October 2000-December 2005 (NCDC DSI-3211)
- b = Belarus update
- C = Environment Canada
- E = European Climate Assessment and Dataset (Klein Tank et al., 2002)
- F = U.S. Fort data
- G = Official Global Climate Observing System (GCOS) or other government-supplied data
- H = High Plains Regional Climate Center real-time data
- I = International collection (non U.S. data received through personal contacts)
- K = U.S. Cooperative Summary of the Day data digitized from paper observer forms (from 2011 to present)
- M = Monthly METAR Extract (additional ASOS data)
- N = Community Collaborative Rain, Hail,and Snow (CoCoRaHS)
- Q = Data from several African countries that had been "quarantined", that is, withheld from public release until permission was granted from the respective meteorological services

- R = NCDC Reference Network Database (Climate Reference Network and Historical Climatology Network-Modernized)
- r = All-Russian Research Institute of Hydrometeorological Information-World Data Center
- S = Global Summary of the Day (NCDC DSI-9618) NOTE: "S" values are derived from hourly synoptic reports exchanged on the Global Telecommunications System (GTS). Daily values derived in this fashion may differ significantly from "true" daily data, particularly for precipitation(i.e., use with caution).
- s = China Meteorological Administration/National Meteorological Information Center/Climate Data Center (http://cdc.cma.gov.cn)
- T = SNOwpack TELemtry (SNOTEL) data obtained from the Western Regional Climate Center
- U = Remote Automatic Weather Station (RAWS) data obtained from the Western Regional Climate Center
- u = Ukraine update
- W = WBAN/ASOS Summary of the Day from NCDC's Integrated Surface Data (ISD).
- X = U.S. First-Order Summary of the Day (NCDC DSI-3210)
- Z = Datzilla official additions or replacements
- z = Uzbekistan update

Table 4 (observation/value)
Note: 9's in a field (e.g.9999) indicate missing data or data that has not been received.
The five core values are:

- PRCP = Precipitation (mm or inches as per user preference, inches to hundredths on Daily Form pdf file)
- SNOW = Snowfall (mm or inches as per user preference, inches to tenths on Daily Form pdf file)
- SNWD = Snow depth (mm or inches as per user preference, inches on Daily Form pdf file)
- TMAX = Maximum temperature (Fahrenheit or Celsius as per user preference, Fahrenheit to tenths on
- Daily Form pdf file
- TMIN = Minimum temperature (Fahrenheit or Celsius as per user preference, Fahrenheit to tenths on
- Daily Form pdf file

The other values are:
- ACMC = Average cloudiness midnight to midnight from 30-second ceilometer data (percent)
- ACMH = Average cloudiness midnight to midnight from manual observations (percent)
- ACSC = Average cloudiness sunrise to sunset from 30-second ceilometer data (percent)
- ACSH = Average cloudiness sunrise to sunset from manual observations (percent)
- AWND = Average daily wind speed (meters per second or miles per hour as per user preference)
- DAEV = Number of days included in the multiday evaporation total (MDEV)
- DAPR = Number of days included in the multiday precipitation total (MDPR)
- DASF = Number of days included in the multiday snowfall total (MDSF)
- DATN = Number of days included in the multiday minimum temperature (MDTN)
- DATX = Number of days included in the multiday maximum temperature (MDTX)
- DAWM = Number of days included in the multiday wind movement (MDWM)
- DWPR = Number of days with non-zero precipitation included in multiday precipitation total (MDPR)
- EVAP = Evaporation of water from evaporation pan (mm or inches as per user preference, or hundredths of inches on Daily Form pdf file)
- FMTM = Time of fastest mile or fastest 1-minute wind (hours and minutes, i.e., HHMM)
- FRGB = Base of frozen ground layer (cm or inches as per user preference)

- FRGT = Top of frozen ground layer (cm or inches as per user preference)
- FRTH = Thickness of frozen ground layer (cm or inches as per user preference)
- GAHT = Difference between river and gauge height (cm or inches as per user preference)
- MDEV = Multiday evaporation total (mm or inches as per user preference; use with DAEV)
- MDPR = Multiday precipitation total (mm or inches as per user preference; use with DAPR and DWPR, if available)
- MDSF = Multiday snowfall total (mm or inches as per user preference)
- MDTN = Multiday minimum temperature (Fahrenheit or Celsius as per user preference ; use with DATN)
- MDTX = Multiday maximum temperature (Fahrenheit or Celsius as per user preference ; use with DATX)
- MDWM = Multiday wind movement (miles or km as per user preference)
- MNPN = Daily minimum temperature of water in an evaporation pan (Fahrenheit or Celsius as per user preference)
- MXPN = Daily maximum temperature of water in an evaporation pan (Fahrenheit or Celsius as per user preference)
- PGTM = Peak gust time (hours and minutes, i.e., HHMM)
- PSUN = Daily percent of possible sunshine (percent)
- SN*# = Minimum soil temperature where * corresponds to a code for ground cover and # corresponds to a code for soil depth (Fahrenheit or Celsius as per user preference)

  Ground cover codes include the following:
  0 = unknown
  1 = grass
  2 = fallow
  3 = bare ground
  4 = brome grass
  5 = sod
  6 = straw mulch
  7 = grass muck
  8 = bare muck

  Depth codes include the following:
  1 = 5 cm
  2 = 10 cm
  3 = 20 cm
  4 = 50 cm
  5 = 100 cm
  6 = 150 cm
  7 = 180 cm

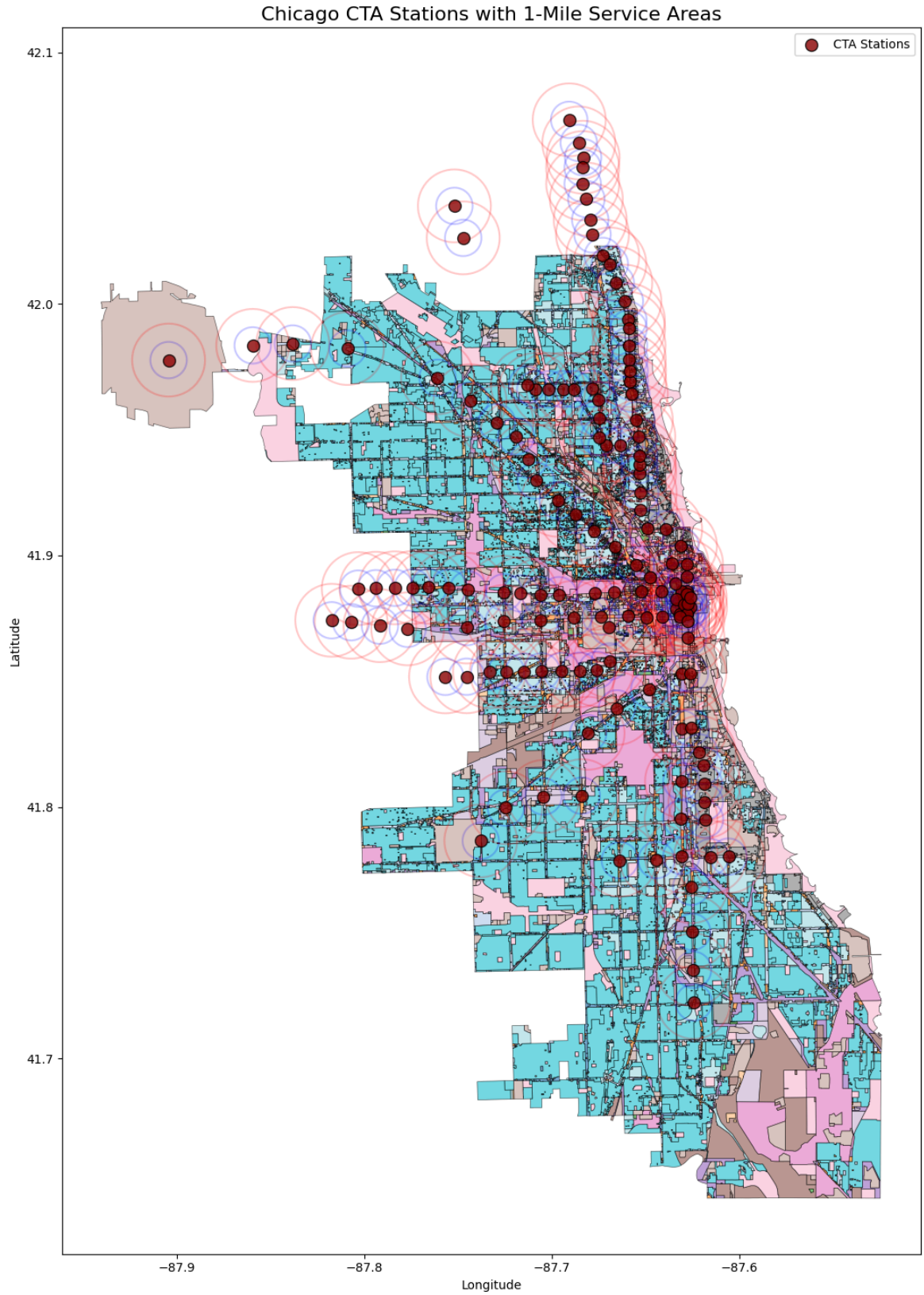
- SX*# = Maximum soil temperature where * corresponds to a code for ground cover and # corresponds to a code for soil depth. See SN*# for depth codes. (Fahrenheit or Celsius as per user preference)
- THIC = Thickness of ice on water (inches or mm as per user preference)
- TOBS = Temperature at the time of observation (Fahrenheit or Celsius as per user preference)
- TSUN = Daily total sunshine (minutes)
- WDF1 = Direction of fastest 1-minute wind (degrees)
- WDF2 = Direction of fastest 2-minute wind (degrees)
- WDF5 = Direction of fastest 5-second wind (degrees)
- WDFG = Direction of peak wind gust (degrees)

- WDFI = Direction of highest instantaneous wind (degrees)
- WDFM = Fastest mile wind direction (degrees)
- WDMV = 24-hour wind movement (km or miles as per user preference, miles on Daily Form pdf file)
- WESD = Water equivalent of snow on the ground (inches or mm as per user preference)
- WESF = Water equivalent of snowfall (inches or mm as per user preference)
- WSF1 = Fastest 1-minute wind speed (miles per hour or meters per second as per user preference)
- WSF2 = Fastest 2-minute wind speed (miles per hour or meters per second as per user preference)
- WSF5 = Fastest 5-second wind speed (miles per hour or meters per second as per user preference)
- WSFG = Peak guest wind speed (miles per hour or meters per second as per user preference)
- WSFI = Highest instantaneous wind speed (miles per hour or meters per second as per user preference)
- WSFM = Fastest mile wind speed (miles per hour or meters per second as per user preference)
- WT** = Weather Type where ** has one of the following values:
    - 01 = Fog, ice fog, or freezing fog (may include heavy fog)
    - 02 = Heavy fog or heaving freezing fog (not always distinguished from fog)
    - 03 = Thunder
    - 04 = Ice pellets, sleet, snow pellets, or small hail
    - 05 = Hail (may include small hail)
    - 06 = Glaze or rime
    - 07 = Dust, volcanic ash, blowing dust, blowing sand, or blowing obstruction
    - 08 = Smoke or haze
    - 09 = Blowing or drifting snow
    - 10 = Tornado, waterspout, or funnel cloud
    - 11 = High or damaging winds
    - 12 = Blowing spray
    - 13 = Mist
    - 14 = Drizzle
    - 15 = Freezing drizzle
    - 16 = Rain (may include freezing rain, drizzle, and freezing drizzle)
    - 17 = Freezing rain
    - 18 = Snow, snow pellets, snow grains, or ice crystals
    - 19 = Unknown source of precipitation
    - 21 = Ground fog
    - 22 = Ice fog or freezing fog

- WVxx = Weather in the Vicinity where “xx” has one of the following values
    - 01 = Fog, ice fog, or freezing fog (may include heavy fog)
    - 03 = Thunder
    - 07 = Ash, dust, sand, or other blowing obstruction
    - 18 = Snow or ice crystals
    - 20 = Rain or snow shower

Additional details are available online at http://www1.ncdc.noaa.gov/pub/data/ghcn/daily/readme.txt (note the readme file does not apply to the Daily Form pdf file or CSV outputs, however values are the same, other than a few differences in units noted in table 4 above).

Map of 'L' Stations and their Radii



Final Features

*Highlighted features are dummy variables
***Italicized features are zoning variables*

Feature Name	Data Type	Example Values	Feature Name	Data Type	Example Values
days_since_2000	int64	8256, 9286, 1998	*commercial_manufacturing_and_employment_district_within_mile*	*float64*	*0.0, 1.0*
post_shutdown	int64	0, 1	*commercial_manufacturing_and_employment_district_within_half_mile*	*float64*	*0.0, 1.0*
season_Spring	int64	0, 1	*downtown_core_district_within_mile*	*float64*	*0.0, 1.0*
season_Summer	int64	0, 1	*downtown_core_district_within_half_mile*	*float64*	*0.0, 1.0*
season_Winter	int64	0, 1	*downtown_residential_district_within_mile*	*float64*	*0.0, 1.0*
year	int32	2024, 2017, 2020	*downtown_residential_district_within_half_mile*	*float64*	*0.0, 1.0*
PRCP	float64	0.0, 0.0, 0.86	*downtown_service_district_within_mile*	*float64*	*0.0, 1.0*
TAVG	float64	75.0, 40.0, 26.0	*downtown_mixed_use_district_within_half_mile*	*float64*	*0.0, 1.0*
SNOW	float64	0.0, 0.0, 0.0	*downtown_mixed_use_district_within_mile*	*float64*	*0.0, 1.0*
SNWD	float64	2.0, 3.9, 0.0	*downtown_mixed_use_district_within_half_mile*	*float64*	*0.0, 1.0*
Fog_Ice_Fog	int64	0, 1	*limited_manufacturing_business_park_district_within_mile*	*float64*	*0.0, 1.0*
Blowing_Snow	int64	0, 1	*limited_manufacturing_business_park_district_within_half_mile*	*float64*	*0.0, 1.0*
latitude	float64	41.84678, 41.854109, 41.961539	*light_industry_district_within_mile*	*float64*	*0.0, 1.0*
longitude	float64	-87.683337, -87.708541, -87.704406	*light_industry_district_within_half_mile*	*float64*	*0.0, 1.0*
ADA	float64	0.0, 1.0	*heavy_industry_district_within_mile*	*float64*	*0.0, 1.0*
RED	float64	0.0, 1.0	*heavy_industry_district_within_half_mile*	*float64*	*0.0, 1.0*
BLUE	float64	0.0, 1.0	*planned_developments_within_mile*	*float64*	*0.0, 1.0*

G	float64	0.0, 1.0	*planned_developments_ within_half_mile*	*float64*	*0.0, 1.0*
BRN	float64	0.0, 1.0	*planned_manufacturing_ district_within_mile*	*float64*	*0.0, 1.0*
P	float64	0.0, 1.0	*planned_manufacturing_ district_within_half_mile*	*float64*	*0.0, 1.0*
Y	float64	0.0, 1.0	*regional_or_community_park _within_mile*	*float64*	*0.0, 1.0*
Pnk	float64	0.0, 1.0	*regional_or_community_park _within_half_mile*	*float64*	*0.0, 1.0*
O	float64	0.0, 1.0	*residential_multi_unit_ district_within_mile*	*float64*	*0.0, 1.0*
N/S	float64	0.0, 1.0	*residential_multi_unit_ district_within_half_mile*	*float64*	*0.0, 1.0*
E/W	float64	0.0, 1.0	*residential_single_unit_ district_within_mile*	*float64*	*0.0, 1.0*
neighborhood_shopping_ district_within_mile	*float64*	*0.0, 1.0*	*residential_single_unit_ district_within_half_mile*	*float64*	*0.0, 1.0*
neighborhood_shopping_ district_within_half_mile	*float64*	*0.0, 1.0*	*residential_two_flat_ townhouse_and_multi_unit_ district_within_mile*	*float64*	*0.0, 1.0*
neighborhood_mixed_use_ district_within_mile	*float64*	*0.0, 1.0*	*residential_two_flat_ townhouse_and_multi_unit_ district_within_half_mile*	*float64*	*0.0, 1.0*
neighborhood_mixed_use_ district_within_half_mile	*float64*	*0.0, 1.0*	*transportation_district_ within_mile*	*float64*	*0.0, 1.0*
community_shopping_ district_within_mile	*float64*	*0.0, 1.0*	*transportation_district_ within_half_mile*	*float64*	*0.0, 1.0*
community_shopping_ district_within_half_mile	*float64*	*0.0, 1.0*	daytype_U	int64	0, 1
neighborhood_commercial_ district_within_mile	*float64*	*0.0, 1.0*	daytype_W	int64	0, 1
neighborhood_commercial_ district_within_half_mile	*float64*	*0.0, 1.0*	rides	int64	6104, 287, 773
motor_vehicle_related_ commercial_district_within _mile	*float64*	*0.0, 1.0*			
motor_vehicle_related_ commercial_district_within _half_mile	*float64*	*0.0, 1.0*			

Decision Tree Boundaries/Rules (Top 5 Levels)

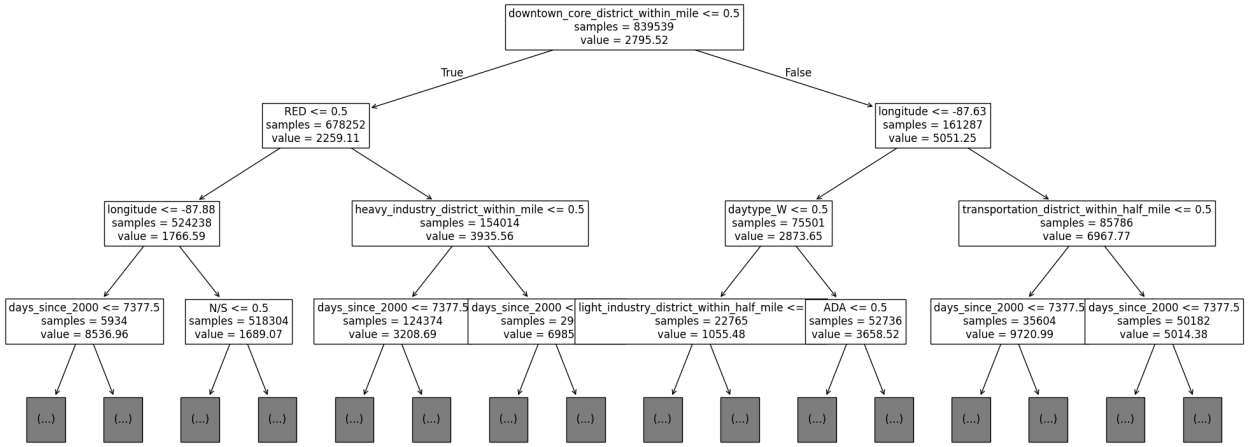
```
|--- downtown_core_district_within_mile <= 0.50
|   |--- RED <= 0.50
|   |   |--- longitude <= -87.88
|   |   |   |--- days_since_2000 <= 7377.50
|   |   |   |   |--- days_since_2000 <= 1961.50
|   |   |   |   |   |--- daytype_W <= 0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- daytype_W >  0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |--- days_since_2000 >  1961.50
|   |   |   |   |   |--- TAVG <= 43.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- TAVG >  43.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |--- days_since_2000 >  7377.50
|   |   |   |   |--- days_since_2000 <= 7816.50
|   |   |   |   |   |--- days_since_2000 <= 7757.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- days_since_2000 >  7757.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |--- days_since_2000 >  7816.50
|   |   |   |   |   |--- days_since_2000 <= 8523.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- days_since_2000 >  8523.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |--- longitude >  -87.88
|   |   |   |--- N/S <= 0.50
|   |   |   |   |--- daytype_W <= 0.50
|   |   |   |   |   |--- latitude <= 41.87
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- latitude >  41.87
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |--- daytype_W >  0.50
|   |   |   |   |   |--- days_since_2000 <= 7378.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- days_since_2000 >  7378.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |--- N/S >  0.50
|   |   |   |   |--- BLUE <= 0.50
|   |   |   |   |   |--- latitude <= 41.79
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- latitude >  41.79
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |--- BLUE >  0.50
|   |   |   |   |   |--- days_since_2000 <= 7377.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- days_since_2000 >  7377.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |--- RED >  0.50
```

```
|   |   |--- heavy_industry_district_within_mile <= 0.50
|   |   |   |--- days_since_2000 <= 7377.50
|   |   |   |   |--- daytype_W <= 0.50
|   |   |   |   |   |--- residential_multi_unit_district_within_half_mile <= 0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- residential_multi_unit_district_within_half_mile >  0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |--- daytype_W >  0.50
|   |   |   |   |   |--- light_industry_district_within_mile <= 0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- light_industry_district_within_mile >  0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |--- days_since_2000 >  7377.50
|   |   |   |   |--- P <= 0.50
|   |   |   |   |   |--- residential_multi_unit_district_within_half_mile <= 0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- residential_multi_unit_district_within_half_mile >  0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |--- P >  0.50
|   |   |   |   |   |--- days_since_2000 <= 7821.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- days_since_2000 >  7821.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |--- heavy_industry_district_within_mile >  0.50
|   |   |   |--- days_since_2000 <= 7377.50
|   |   |   |   |--- downtown_service_district_within_mile <= 0.50
|   |   |   |   |   |--- daytype_W <= 0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- daytype_W >  0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |--- downtown_service_district_within_mile >  0.50
|   |   |   |   |   |--- latitude <= 41.91
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- latitude >  41.91
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |--- days_since_2000 >  7377.50
|   |   |   |   |--- days_since_2000 <= 7829.50
|   |   |   |   |   |--- latitude <= 41.81
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- latitude >  41.81
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |--- days_since_2000 >  7829.50
|   |   |   |   |   |--- BRN <= 0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- BRN >  0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|--- downtown_core_district_within_mile >  0.50
|   |--- longitude <= -87.63
|   |   |--- daytype_W <= 0.50
|   |   |   |--- light_industry_district_within_half_mile <= 0.50
|   |   |   |   |--- latitude <= 41.88
```

```
| | | | | |--- P <= 0.50
| | | | | | |--- truncated branch of depth 5
| | | | | |--- P >  0.50
| | | | | | |--- truncated branch of depth 5
| | | | |--- latitude >  41.88
| | | | | |--- days_since_2000 <= 1985.00
| | | | | | |--- truncated branch of depth 5
| | | | | |--- days_since_2000 >  1985.00
| | | | | | |--- truncated branch of depth 5
| | | |--- light_industry_district_within_half_mile >  0.50
| | | | |--- year <= 2009.00
| | | | | |--- commercial_manufacturing_and_employment_district_within_half_mile <= 0.50
| | | | | | |--- truncated branch of depth 5
| | | | | |--- commercial_manufacturing_and_employment_district_within_half_mile >  0.50
| | | | | | |--- truncated branch of depth 5
| | | | |--- year >  2009.00
| | | | | |--- days_since_2000 <= 7266.50
| | | | | | |--- truncated branch of depth 5
| | | | | |--- days_since_2000 >  7266.50
| | | | | | |--- truncated branch of depth 5
| | |--- daytype_W >  0.50
| | | |--- ADA <= 0.50
| | | | |--- post_shutdown <= 0.50
| | | | | |--- transportation_district_within_mile <= 0.50
| | | | | | |--- truncated branch of depth 5
| | | | | |--- transportation_district_within_mile >  0.50
| | | | | | |--- truncated branch of depth 5
| | | | |--- post_shutdown >  0.50
| | | | | |--- days_since_2000 <= 7863.50
| | | | | | |--- truncated branch of depth 5
| | | | | |--- days_since_2000 >  7863.50
| | | | | | |--- truncated branch of depth 5
| | | |--- ADA >  0.50
| | | | | |--- days_since_2000 <= 7376.50
| | | | | |--- residential_single_unit_district_within_mile <= 0.50
| | | | | | |--- truncated branch of depth 5
| | | | | |--- residential_single_unit_district_within_mile >  0.50
| | | | | | |--- truncated branch of depth 5
| | | | |--- days_since_2000 >  7376.50
| | | | | |--- days_since_2000 <= 7870.50
| | | | | | |--- truncated branch of depth 5
| | | | | |--- days_since_2000 >  7870.50
| | | | | | |--- truncated branch of depth 5
| |--- longitude >  -87.63
| | |--- transportation_district_within_half_mile <= 0.50
| | | |--- days_since_2000 <= 7377.50
| | | | |--- daytype_W <= 0.50
| | | | | |--- RED <= 0.50
| | | | | | |--- truncated branch of depth 5
| | | | | |--- RED >  0.50
| | | | | | |--- truncated branch of depth 5
```

```
|   |   |   |   |--- daytype_W >  0.50
|   |   |   |   |   |--- ADA <= 0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- ADA >  0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |--- days_since_2000 >  7377.50
|   |   |   |   |--- days_since_2000 <= 7828.50
|   |   |   |   |   |--- days_since_2000 <= 7736.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- days_since_2000 >  7736.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |--- days_since_2000 >  7828.50
|   |   |   |   |   |--- daytype_U <= 0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- daytype_U >  0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |--- transportation_district_within_half_mile >  0.50
|   |   |   |--- days_since_2000 <= 7377.50
|   |   |   |   |--- daytype_W <= 0.50
|   |   |   |   |   |--- neighborhood_mixed_use_district_within_mile <= 0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- neighborhood_mixed_use_district_within_mile >  0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |--- daytype_W >  0.50
|   |   |   |   |   |--- N/S <= 0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- N/S >  0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |--- days_since_2000 >  7377.50
|   |   |   |   |--- days_since_2000 <= 7856.50
|   |   |   |   |   |--- residential_two_flat_townhouse_and_multi_unit_district_within_mile <= 0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- residential_two_flat_townhouse_and_multi_unit_district_within_mile >  0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |--- days_since_2000 >  7856.50
|   |   |   |   |   |--- G <= 0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
|   |   |   |   |   |--- G >  0.50
|   |   |   |   |   |   |--- truncated branch of depth 5
```

Diagram of Top 3 Levels



Top 20 Features in Each Principal Component

Principal Component 1

	PC1
community_shopping_district_within_half_mile	0.225829
residential_two_flat_townhouse_and_multi_unit_district_within_half_mile	0.222301
neighborhood_shopping_district_within_half_mile	0.218255
neighborhood_commercial_district_within_half_mile	0.217406
neighborhood_shopping_district_within_mile	0.215226
neighborhood_mixed_use_district_within_mile	0.212420
residential_two_flat_townhouse_and_multi_unit_district_within_mile	0.209781
residential_single_unit_district_within_mile	0.204785
limited_manufacturing_business_park_district_within_mile	0.200990
motor_vehicle_related_commercial_district_within_mile	0.196752
community_shopping_district_within_mile	0.196578
residential_single_unit_district_within_half_mile	0.190622
neighborhood_mixed_use_district_within_half_mile	0.190012
motor_vehicle_related_commercial_district_within_half_mile	0.180096
neighborhood_commercial_district_within_mile	0.179477
regional_or_community_park_within_mile	0.162072
limited_manufacturing_business_park_district_within_half_mile	0.160007
planned_developments_within_half_mile	0.158185
regional_or_community_park_within_half_mile	0.157578
planned_developments_within_mile	0.154657

Principal Component 2

	PC2
downtown_mixed_use_district_within_mile	0.266977
downtown_residential_district_within_mile	0.265973
downtown_service_district_within_mile	0.265913
downtown_mixed_use_district_within_half_mile	0.264126
downtown_core_district_within_mile	0.250356
downtown_core_district_within_half_mile	0.222678
downtown_residential_district_within_half_mile	0.220747
downtown_service_district_within_half_mile	0.217538
longitude	0.189401
light_industry_district_within_mile	0.181008
planned_developments_within_half_mile	0.171607
regional_or_community_park_within_mile	0.158281
planned_developments_within_mile	0.155269
neighborhood_commercial_district_within_mile	0.143958
transportation_district_within_mile	0.139598
commercial_manufacturing_and_employment_district_within_mile	0.129360
residential_single_unit_district_within_half_mile	-0.105397
O	0.104924
latitude	-0.104464
motor_vehicle_related_commercial_district_within_mile	0.099500

Principal Component 3

	PC3
E/W	0.342770
N/S	-0.338330
RED	-0.271549
transportation_district_within_half_mile	-0.218482
light_industry_district_within_half_mile	0.197994
transportation_district_within_mile	-0.191105
Pnk	0.190641
longitude	-0.157651
latitude	-0.146697
MAP_ID	-0.139093
station_id	-0.139093
G	0.133338
limited_manufacturing_business_park_district_within_half_mile	0.123678
regional_or_community_park_within_half_mile	-0.111226
heavy_industry_district_within_mile	0.106171
heavy_industry_district_within_half_mile	0.095735
BLUE	0.086969
commercial_manufacturing_and_employment_district_within_half_mile	0.082424
downtown_service_district_within_half_mile	0.074663
commercial_manufacturing_and_employment_district_within_mile	0.073889